

Study Questions: Systems

1. Define **Systems** in your own words, specifically as it applies to Middle School.
2. How does the Free Energy Principle constrain our understanding of Systems?
3. Contrast the Classical view of Systems with the Active Inference view.
4. Name three systems you are part of right now (besides your family). For each one, list two parts and explain how they are connected.
5. Your friend group is a system. What happens to the whole group when two people in it start arguing? Trace at least two ripple effects.
6. Think about your school cafeteria as a system. What are the inputs (things that go in), outputs (things that come out), and at least one feedback loop?
7. If your school decided to ban phones during lunch, map out three ripple effects that change would cause through the school system.
8. What is a **boundary** in a system? Give an example of a boundary in your life that is clear and one that is fuzzy.
9. A basketball team and a pile of five random people both have five humans. Why is the team a system but the random group is not? What is the difference?
10. Imagine you could remove one part of your morning routine system. Which part would cause the biggest ripple effect if it disappeared, and why?
11. How is a group chat like a system? Identify the parts, connections, and what the system does that individual members cannot do alone.
12. Compare your family system to your school system. What do they have in common? How are the boundaries different?
13. Give an example of a **positive** ripple effect in a system you belong to, where one small change made many things better.

14. Why is it important to think about systems before making a big decision? Use the example of posting something on social media to explain.
15. Draw or describe a system that includes at least four parts and three connections. Label the boundary. It can be real or made up.
16. Your teacher is absent for a week. Trace how this one change ripples through at least three other parts of the school system.
17. Can a single person be a system? Why or why not? Use what you learned about parts, connections, and purpose to argue your answer.
18. Explain why the statement "you can't change just one thing" is true in a system. Use a specific example from your daily life.
19. How would understanding systems help you solve a conflict between two friends? What would you look at that most people miss?
20. Design a brand-new system for organizing a school event. Identify at least three parts, how they connect, and what could go wrong if one part fails.